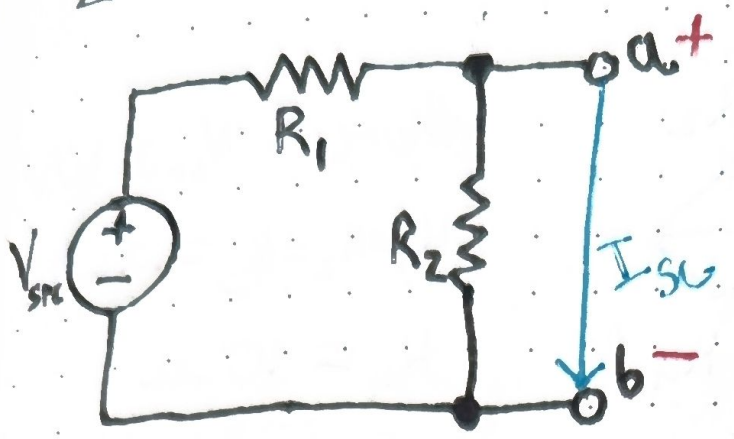
10 Ω

$$= \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

$= 0.15A$

$I_A = 0.11A$

250922 Thévenin/Norton



Open-Circuit:

$V_{ab} = V_{oc}$

Using Volt-divider →

$\rightarrow V_{ab} = \frac{R_2}{R_1 + R_2} V_{src}$

Short-Circuit:

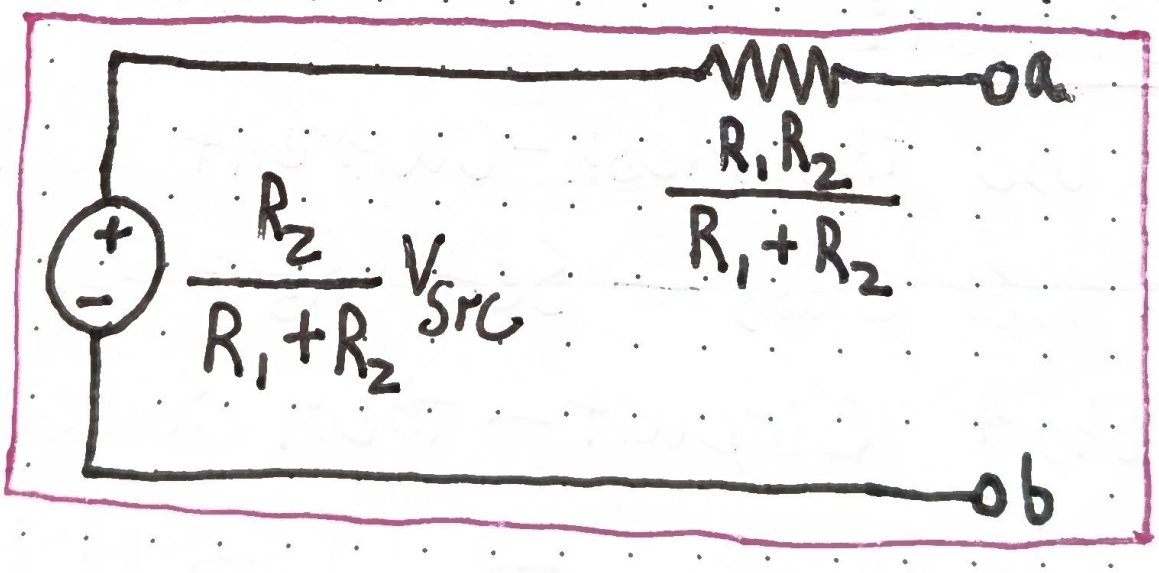
Using Ohm's law → $I_{sc} = V_{src} / R_1$

(R_2 is ignored since its shorted)

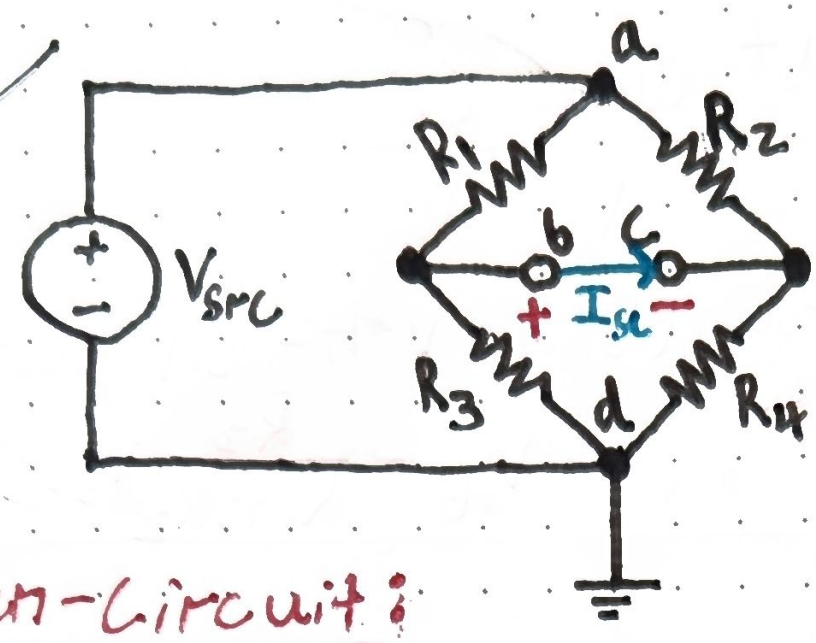
Thévenin Equiv: $V_{Th} = V_{oc} = V_{ab}$

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \left(\frac{R_2}{R_1 + R_2} V_{src}\right)\left(\frac{R_1}{V_{src}}\right) = \frac{R_1 R_2}{R_1 + R_2}$$

$\therefore$



ex/



Where: $V_{src} = 10V$

$R_1 = R_2 = R_3 = 100\,\Omega$

$R_4 = 10\,\Omega$

Open-Circuit:

$$V_{oc} = V_{bc} = V_{Th} = V_b - V_c$$

Find w/ Volt-divider

$$V_b = \frac{R_3}{R_1 + R_3} V_{src} \qquad V_c = \frac{R_4}{R_2 + R_4} V_{src}$$

$$\Rightarrow V_{bc} = \left(\frac{R_3}{R_1+R_3} - \frac{R_4}{R_2+R_4}\right) V_{src} \Rightarrow \boxed{V_{oc} = 4.1V}$$

Short-Circuit:

Let's use the Mesh-Current Solution from last Class — Set $R_5 = 0\,\Omega$ for a Short Circuit — then Solve for I_5.

$$\begin{bmatrix} 0 \\ 0 \\ V_{src} \end{bmatrix} = \begin{bmatrix} R_1+R_2+R_5 & -R_5 & -R_1 \\ -R_5 & R_3+R_4+R_5 & -R_3 \\ -R_1 & -R_3 & R_1+R_3 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ \\ \end{bmatrix}$$

$\Rightarrow I_A = 0$

$\Rightarrow I_5 =$

Thévenin

$$R_{Th} = \frac{V_{oc}}{I_5}$$

$\therefore$

Norton

$\Rightarrow I_N$

$R_N =$

$\therefore$ 69mA

$c : V_{src} = 10V$

$R_2 = R_3 = 100\,\Omega$

$= 10\,\Omega$

olt-divider

$_c = 4.1V$

olution

2 for

for I_5.

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 200 & 0 & -100 \\ 0 & 110 & -100 \\ -100 & -100 & 200 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

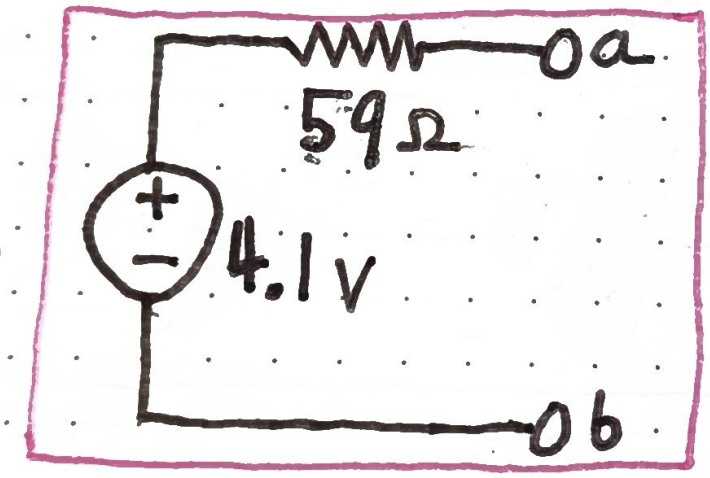
$\Rightarrow I_A = 0.085A \quad I_B = 0.15A \quad I_C = 0.17A$

$\Rightarrow I_5 = I_B - I_A = \boxed{0.069A = I_{sc}}$

Thévenin Equiv:

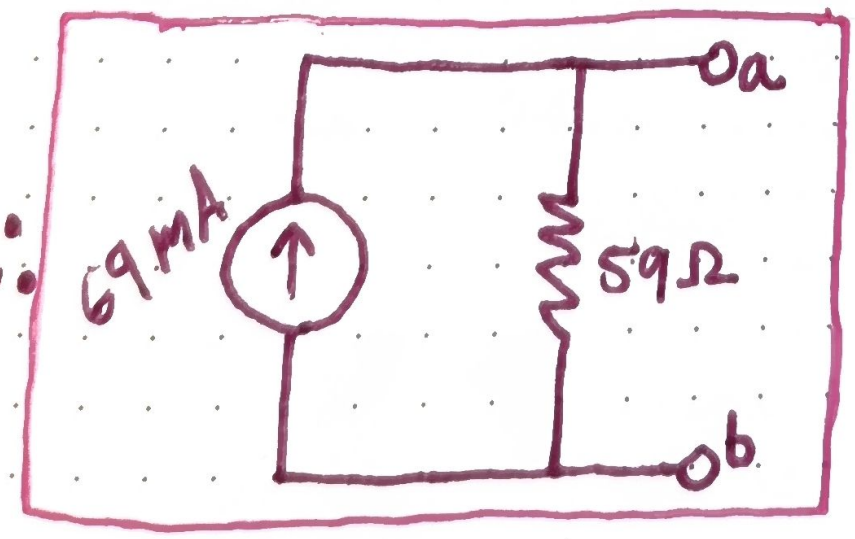
$$R_{Th} = \frac{V_{oc}}{I_{sc}} = 4.1/0.069 = \boxed{59\,\Omega = R_{Th}}$$

$\therefore$



Norton Equiv: $I_N = V_{Th}/R_{Th} = I_{sc}$

$\Rightarrow I_N = 0.069\,A$

$R_N = R_{Th} \Rightarrow R_N = 59\,\Omega$

$\therefore$